Mapping Flood Damage from Space — Report

Multimodal AI Hackathon 2026, Track B

1. Problem

When roads, bridges and phone lines are gone, rescuers need to know where the damage is and who is cut off. Our system takes **an area and a flood date** and, from free satellite data only, answers: **(1) where did the flood hit, (2) what is damaged, (3) who is cut off.** It runs end-to-end from raw satellite data for any area and date (nothing is hard-wired for Nepal). Case study: the 26 August 2026 Bhote Koshi–Trishuli flood. This is an educational prototype, not an operational tool.

2. Data (allowed sources only)

Sentinel-1 radar and Sentinel-2 optical images (Copernicus Data Space / Sentinel Hub, AWS Open Data), Copernicus DEM GLO-30, OpenStreetMap **as it was before the event** (ohsome API snapshot at 00:00 UTC the day before; never Overpass or live OSM), and the training datasets Kuro Siwo and Sen1Floods11. **Copernicus EMS, UNOSAT or any published damage map is never an input** — not for training, thresholds or pipeline; EMSR927 appears only in `validation/` and in Section 8 to check finished results. Thresholds are set from training data or literature, never tuned on EMSR927.

3. System overview

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area + date → download (area only): S1 before/after, same orbit track, both pass directions; S2 if clear; DEM; OSM snapshot
 → preprocessing (10 m UTM grid, layover/shadow mask, slope, HAND)
 → flood water (AI model)   → debris / mud / flooded vegetation (change test)
 → final flood map → damage (buildings, roads, bridges, facilities)
 → road network before vs after → cut-off settlements, repair order
 → stats.json → one-page situation report + map overlays → dashboard
```

Steps communicate through files in one run folder; `stats.json` is the single source of every number in the dashboard and report. Key rules: radar change detection uses **pre and post from the same relative orbit**; the “before” image is checked so that it is not already flooded (water-fraction time series, earlier image used if needed); the main track is the earliest post-event image unless the opposite direction sees the valley floor clearly better. If no usable pair exists, the system says *insufficient data*; if nothing is found it says *no significant flooding* **together with coverage** (what could be seen).

4. AI component: flood segmentation model

Model. U-Net with ResNet34 encoder (ImageNet initialisation, allowed by the organisers), 7 input channels: before/after VV and VH backscatter (dB), their differences, and terrain slope; 3 classes (no water / permanent water / flood). Trained on Kuro Siwo (Bountos et al., 2024) plus Sen1Floods11 (Bonafilia et al., 2020) with Lovász loss; probabilities calibrated (temperature 0.91). Sentinel-1 is processed to match Kuro Siwo (sigma0, Lee 7×7 + 3×3 mean): measured bias 0.00 dB, speckle 0.96× of Kuro Siwo on 28 tiles. Runs on a CPU (~30 s per 30×30 km per track). An optional optical U-Net (ResNet18, Sentinel-2) adds water where the sky is clear. Post-processing: no water on slopes > 15° or HAND > 30 m, radar shadow checked from both directions, patches < 0.1 ha removed; *Standard* and *Sensitive* modes (probability threshold 0.25).

Results (radar model v0.5; events never seen in training; pixels pooled):

Test	Water class	Precision	Recall	F1	IoU
Kuro Siwo test (10 events, 15 896 tiles)	flood	0.865	0.731	**0.792**	0.656
Kuro Siwo test	permanent water	0.816	0.754	0.784	0.644
Kuro Siwo test	water total	0.893	0.779	0.832	0.713
Kuro Siwo test, mountain tiles (relief > 200 m)	flood	–	–	0.52	–
Sen1Floods11 Bolivia (held out), no “before” image	water total	0.958	0.109	**0.195**	0.108
Sen1Floods11 Bolivia, “before” = “after” (no change)	water total	0.889	0.495	0.636	–
Optical model, Sen1Floods11 Bolivia (held out)	water	0.899	0.872	**0.885**	0.794

For reference, the best model in the Kuro Siwo paper reaches F1 0.801 on these events. We report the Bolivia numbers **honestly**: the radar model learned to rely on change between before and after, so with no “before” image it rarely reports water (F1 0.20); Bolivian water is also bright in radar (median −16 dB). That setting is not how the pipeline runs (it always has a “before” image), but it shows the model is a **change detector, not a general water detector**.

Differences between model variants after round 3 are within seed noise (val score ±0.007); the only clearly repeatable gain came from the terrain-slope input.

Whole pipeline on 14 independent events (`validation/results/multievent/`). The table above scores the model on tiles; the judges' question is whether the *whole system* works anywhere from raw data. We ran it end to end (download → final map, Sensitive mode) on 13 floods on five continents and one dry-season no-flood control, all with references that are **not** EMS/UNOSAT: Kuro Siwo test hand labels (Australia, Bihar, Chad, Evros, Honduras, Malawi, Missouri), Sen1Floods11 (Bolivia) and, for mountain events where no dataset exists, our own Sentinel-2 water labels (Teesta, Chamoli, Swat, Kathmandu, Baghlan). Water-label events are scored on water only (debris is not in such a label). Every method change was accepted only if the mean improved, no event lost more than 0.01 F1 and the control gained no false area.

events	recall	precision	F1
9 original floods (mean)	0.86	0.88	**0.87**
12 floods incl. 3 added mountain floods (mean)	0.85	0.81	**0.81**
lowland/large rivers: Australia, Bihar, Bolivia, Evros, Malawi, Missouri, Swat	0.85–0.99	0.81–0.99	0.87–0.94
Honduras / Teesta / Chad	0.83 / 0.64 / 0.67	0.89 / 0.90 / 0.79	0.86 / 0.75 / 0.72
Baghlan / Kathmandu (flash floods, water already drained)	0.68 / 0.86	0.68 / 0.29	0.68 / 0.43
Chamoli 2021 (debris flow; 0.04 km² of water in label, not in the means)	0.06	0.48	0.11

Large river floods are found reliably; narrow mountain channels and flash floods are the weak point. Kathmandu and Baghlan had only 0.10 / 0.35 km² of water left on the label date, so a fraction of a km² of extra water halves precision; their debris (1.5 km²) cannot be checked against a water label. Kathmandu also showed a real error, fixed generically: an airport runway (smooth asphalt is dark in every radar pass) was called permanent water; it is now made dry where both clear Sentinel-2 images see dry land and the year-long radar record agrees. A full run takes 2–7 min for a ~9 km window on a laptop CPU. Caveats: Sentinel-2 labels can share the image the pipeline uses (Baghlan), and the method changes were chosen on this same set.

Fresh test, pre-registered, never-seen scenes. Because the method was tuned on the set above, we then fixed 7 new floods (Uttarakhand / West Nepal, Oct 2021) and a dry-season control *before* running anything and changed nothing afterwards (Sentinel-2 water labels, water only). The full system reaches mean F1 **0.72** (pooled 0.76), but it is partly circular here: its optical "after" image is the label image. Radar alone, which is independent of the label, reaches mean F1 **0.45** (pooled 0.65): 0.81 / 0.72 on the large Karnali and Mahakali rivers, but 0.00–0.28 on narrow or braided channels (its pass was also 2–5 days before the label, when water was higher). False alarm on the control: 0.28 % of dry land. Conclusion: reliable on large rivers anywhere; in narrow mountain valleys a clear Sentinel-2 image is needed.

5. Debris, mud and other surface change

Mud, rock and flooded vegetation/buildings do not look like open water in radar: debris and wet mud often darken only slightly, flooded vegetation and towns **brighten** (double bounce). A separate statistical step compares each pixel with its own history (up to 4 earlier passes of the same track), combines both pass directions, grows regions from strong seeds, and labels subtypes (debris, mud, erosion, flooded vegetation, flooded / damaged buildings, unknown) with a confidence level using Sentinel-2 and OSM building/land-use when available. A **self-test on a flood-free image pair** estimates how much of the result may be false; a weather check (change over the surrounding slopes) rejects rain and wet snow. Measured on the Trishuli valley floor: 6.3 % darkening vs 0.4 % brightening; on Opava 2024: 0.5 % / 13.6 %. Tested on a synthetic valley (all subtypes ≥ 90 %, no false pixels elsewhere); real-data thresholds come from literature on floods and landslides, **not measured on Himalayan debris flows**. Results are therefore shown as indicative, with confidence.

6. Damage

Per object, not per village: a building is *affected* if its outline plus a 10 m margin overlaps the flood map, *likely destroyed* if debris/erosion lies under it, *possibly affected* if only the margin or low confidence. Roads are *blocked* at ≥ 17 % of length or ≥ 30 m continuous in the flood zone (stricter rules cut false closures from 16 % to 5 %, Kasmalkar 2020). Bridges: water under a bridge is not evidence of damage; approaches and debris are tested → *blocked / suspect / passable*. Counts are given as a range (certain only … including "possible"), like UNOSAT's "potentially affected". No population estimates (only OSM `population` if present).

7. Who is cut off

We build our own road graph from the pre-event OSM snapshot (networkx; OSMnx would use live OSM). Post-event graph = pre-event graph without blocked road pieces and blocked bridges; a settlement is *connected*, *degraded* (detour > 2×), *possibly cut off* (cut only if suspect bridges/possible blockages are counted), *cut off*, or *no road before the event*. Destinations: hospitals, towns and the edge of the downloaded network. Output includes detour %, walking time, blocking road pieces, a five-level impact grade per settlement with reasons, an urgency ranking, and a **stepwise repair plan** (which road section reconnects the most buildings per repaired section). Helicopter landing-zone candidates from slope in the DEM are always marked *unverified*. Bonus: a click on the map traces the flood path downstream (D8 flow

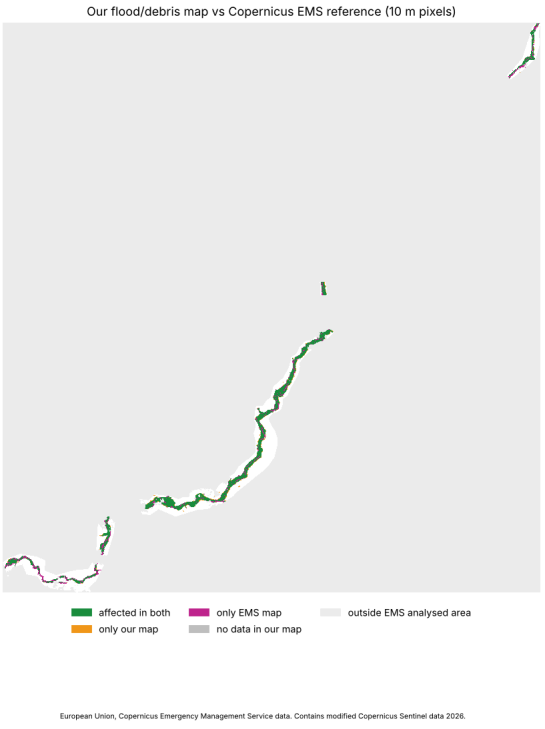
direction on the DEM) and lists settlements, bridges and facilities along it, ordered downstream — no hydraulics, no arrival times.

8. Case study: Trishuli flood vs EMSR927

Run `trishuli_now`: area 85.0–85.4 °E, 27.8–28.2 °N (~2 000 km²), event date 26 Aug 2026, post-event Sentinel-1 image 28 Aug; EMS images 27 Aug (Phosretar 31 Aug). EMS products of 4 areas (Syapru Besi, Timure, Bidur, Phosretar); the other two were cancelled because of clouds. **EMS was used only for this comparison**, after the run, never as input or for tuning (European Union, Copernicus Emergency Management Service data). Scored inside the EMS analysed areas where our map has data (40.4 km²).

check	result
Affected area (our new water + debris vs EMS observed event, 10 m pixels)	precision **0.92**, recall **0.78**, F1 **0.85**, IoU 0.73 (ours 9.97 km², EMS 11.66 km²)
EMS damaged buildings (3 376) inside our affected area (±10 m)	**79 %**; destroyed/damaged only: **90 %**
EMS damaged roads (64 km) near roads we mark blocked	**85 %** (91 % incl. “possibly blocked”); false alarms: 23 % of EMS “no visible damage” roads
EMS destroyed/damaged bridges matched to OSM (14 of 25)	**93 %** marked blocked or suspect

Whole area: new water 9.8 km², debris/mud 79.6 km², 7 400–9 300 buildings affected (959 likely destroyed), 480–1 170 km of road blocked or possibly blocked, 22 bridges blocked + 14 suspect; **45 settlements cut off, 73 possibly cut off**. Run time 24 min (download 6 min, debris step 10 min) and ~2 200 of 10 000 free monthly processing units. Inside the EMS footprint our map has 5.1 km² water and 4.1 km² debris — EMS maps one class (mass movement), so only the union is scored. The overall confidence is *low* (72 % of affected pixels low confidence): steep valley, radar shadow, and debris thresholds from literature. Caveats: 10 m pixels vs sub-metre hand mapping; image dates differ; EMS lists only damaged buildings, so building precision cannot be computed (396 of our 1 979 hit buildings in the area are not in EMS: false alarms or EMS omissions); only small parts of the valley were mapped by EMS.



Our affected area vs EMSR927 inside the EMS analysed areas (European Union, Copernicus Emergency Management Service data).

9. Limitations (what the system cannot do)

- **Revisit time.** Sentinel-1/2 pass an area only every few days (about 6–12 days per track). The system maps what was there on the image dates; it **cannot warn of a sudden event such as a glacier collapse minutes ahead** and may miss short-lived water. A very recent flood may have no “after” image yet; the app then says so and when a better view is expected.
- **Radar shadow and layover** in steep valleys hide part of the ground. Two pass directions help but do not cover everything; the data-quality panel states what share was visible.
- **Narrow, fast mountain rivers** look like land to radar (rough water, channel 2–6 pixels wide): mountain recall 0.06–0.95, weakest for debris flows (Chamoli); flash-flood water is often gone before the next pass.
- **Clouds** (monsoon) limit optical use; then only radar is used.
- **Debris vs wet soil.** Debris/mud detection is heuristic, with literature thresholds not measured on Himalayan debris flows; C-band does not penetrate dense forest; streets aligned with the radar look direction hide urban flooding.

- **The model is a change detector** (see Bolivia result) and is weaker on mountain terrain than on plains (F1 0.52 vs 0.79). Differences between variants are within noise.
- **OpenStreetMap is incomplete** in remote areas (roads, bridges, buildings missing) and its pre-event state is all we know. Footpaths and footbridges are not part of the road network, so a village reachable only by a path is “no road before the event”.
- **10 m pixels:** a “hit” building means flood on or beside it, not a damage grade; no depth or flow velocity.
- **No population estimates**; landing-zone candidates are from the DEM only (trees, wires and obstacles are unknown).
- **No ground truth** was used to tune thresholds; reference comparison only at the end (Section 8).
- **“No significant flooding”** means nothing was found in the available data, not that nothing happened.
- Needs a free Copernicus account and internet (a shared ohsome key is built in; an own free key is optional); about 30×30 km is the recommended area size (a time and quota choice, the app only warns above it).

10. Responsibility

The disaster is recent and many families are still affected. We use no images of victims. Results are a first screening from satellite data, unverified on the ground, and must not be the sole basis of rescue decisions.

11. AI assistance

Code, documentation and this report were developed with **Claude Code** (Anthropic); the team reviewed and can explain every part. Details per module are in the README (section *AI assistance*).

12. Attributions and citations

Contains modified Copernicus Sentinel data 2026. Produced using Copernicus WorldDEM-30 © DLR e.V. 2010–2014 and © Airbus Defence and Space GmbH 2014–2018 provided under COPERNICUS by the European Union and ESA; all rights reserved. © OpenStreetMap contributors (ODbL). Training data: Bountos, N. I. et al. (2024), Kuro Siwo, NeurIPS Datasets and Benchmarks; Bonafilia, D. et al. (2020), Sen1Floods11, CVPR Workshops. Validation only: European Union, Copernicus Emergency Management Service data (EMSR927).